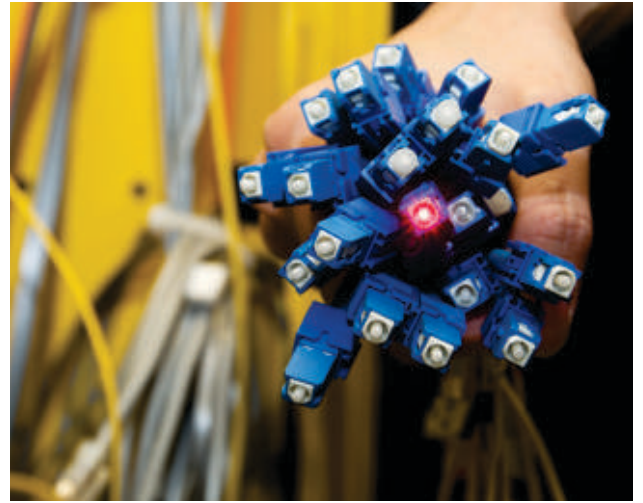




Product Catalog

ABOUT US

CCL Optoelectronics Provides Cutting Edge Fiber Optic Products & Solutions

- We supply to all the leading Telecom (3G, 4G, FTTx) Companies in India
- Our products also have Broadband, Enterprise, Smart Cities, Defence, Railways, Security & Surveillance applications
- In business since 2000
- High Quality Products (ISO 9001:2015 Certified)
- Fast Turnaround; Low Lead Time
- Cost Effective Solutions
- North American Office in Detroit Metropolitan area



Our Vision

A world where every home or business is connected to a fiber optic line, delivering instant video, voice and data.

Our Mission

CCL Optoelectronics is committed to creating an extraordinary customer experience by delivering high quality, innovative connectivity solutions.

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SPLICE JOINT CLOSURES

Our Splice Joint Closures are easy to use and excellent in sealing fiber management & have flexibility in design. Building on renovation & industry leadership. CCL Opto's closures enable faster and future proof network deployment designed to suit various types of cable constructions. Splice joint closures can be deployed in any environment – aerial, pedestal, handhole & manhole. These joint closures have established themselves as industry benchmarks for quality and reliability.

Applications:

- Used to store and secure splice.
- Used for straight through and branching joint applications.
- Suitable for aerial, underground & wall mounting applications

Features:

- Weather, UV and Chemical Resistant and temperature tested for -40 °C to +65 °C, hence can be used in all environments - aerial, underground or buried.
- IP68 sealing protection ensures all weather protection.
- Zero additional losses when optical fibers are routed in splice trays during installation and maintenance.
- Easy installation which requires no specialized tools and minimal training results in cost savings.
- Our rugged design ensures long term reliability.
- Our unique clamp assembly prevents cable movement with temperature changes.
- Splicing trays ensure control on cable bending and the lids provide cable protection.
- Transport tubes for fibers prevent breakage at the tray entry during installation and maintenance.
- The mechanical dome type JC has a re-enterable design which makes it our most economical and future ready option.

Heat Shrink Type

Sr. No	Description	Unit	Particulars
1	Dimension (L x D)	mm	540 x 225
2	Weight	Kg	< 2.8 Kg
3	Cable sealing		Heat shrink sleeves
4	Construction material		
a)	Dome		PPCP
b)	Base		PPCP
c)	Tray		ABS
d)	O-Ring		Neoprene rubber
e)	Clamp		Glass filled nylon
5	Type		Butt type
6	Round port entries for cable	Nos.	4
7	Oval port entries for cable	Nos.	1
8	Total no. of optical splice	Nos.	144
9	Max. no. of cable entries	Nos.	6
10	Max. no. of splice trays	Nos.	6/8
11	No. of splice per tray	Nos.	12/24



Mechanical Type

Sr. No	Description	Unit	Particulars
1	Dimension (L x D)	mm	570 x 225
2	Weight	Kg	< 3.5 Kg
3	Cable sealing		Rubber compression
4	Construction material		
a)	Dome		PPCP
b)	Base		PPCP
c)	Tray		ABS
d)	O-Ring		Neoprene rubber
e)	Clamp		Glass filled nylon
5	Type		Butt type
6	Round port entries for cable	Nos.	4
7	Oval port entries for cable	Nos.	1
8	Total no. of optical splice	Nos.	144
9	Max. no. of cable entries	Nos.	6
10	Max. no. of splice trays	Nos.	6/8
11	No. of splice per tray	Nos.	12/24



Flat Type Splice JC

CCL Opto's Flat Closure is compact, lightweight and its construction is robust and rigid. It is designed to suite all conditions. The application of this closure is to secure fusion splicing, provision of fixing splitting and its splices, and storage of bare and loose tube fibers.

This closure is designed to suite for all types of cable constructions and can be used for straight joint, branch joint, midspan of the optical cables, splitting and distribution connectivity in Telecom and FTTx network.

Applications:

- Used for store and secure splices.
- Used for straight through and branching joint applications.
- Suitable for aerial, underground & wall mounting applications.



Features:

Feature	Benefit
Sealing Protection	IP 68
Zero Additional Loss	No additional loss when optic fibers are routed in the splice trays during installation and maintenance
Unique Strength Member Clamp Assembly	Prevents cable sheath movement with temperature changes
No Specialized Tools Needed	Easy installation
Resistance to UV and Temperature Fluctuation	Can be deployed in all environments: buried, underground and aerial
Bend Controls in Splicing tTray	Guides and protects the pigtail cable and ensures controlled cable bending
Rugged Construction	Long term reliability
Easily Re-Enterable Design	Economical and future ready
Minimal Training, No Specialize Tools Needed	Cost savings, easy installation
Splicing Tray Lid	Protects the splice fibers from coming out of the trays
Transport Tubes for Fibers	Prevents fiber breakage at the tray entry during installation & maintenance
Pressure Valve	Provision to test sealing during operation and maintenance

Specifications:

Sr. No.	Description	Unit	Particulars
1	Dimension (L X W X H)	mm	410 X 280 X 157
2	Weight	Kg	< 3 Kg
3	Cable Sealing	-	Rubber Compression
4	Construction Material	-	NA
a)	Dome	-	PPCP
b)	Base	-	PPCP
c)	Tray	-	ABS
d)	O-Ring	-	EPDM Rubber
e)	Clamp	-	Glass Filled Nylon
5	Type	-	Butt Type
6	Round port entry for cable 9-13 mm (Max)	Nos.	8
7	Oval port entry for cable 12-14 mm (Max)	Nos.	1
8	Total no. of Optical Splice	Nos.	96
9	Degree of protection	-	IP 68
10	Max. no. of cable entries	Nos.	10
11	Max. no. of Splice+Splitter trays	Nos.	12
12	No. of splice per tray	Nos.	12
13	No. of Splitter per tray L x W xH (55~65x7x4)	Nos.	1

FIBER MANAGEMENT SYSTEMS (FMS)

Fiber Management Systems (FMS) provide cross-connect & inter-connect between incoming fiber cables and fiber optic equipment. They offer a modular system to suit various connector options. Our FMS is a CRCA powder coated shelf with provision for individual fiber splicing and termination for distribution. FMS are available from 6 to 288 fiber configurations and can be customized as per customer requirements.

Applications:

- Telecom network
- FTTx deployments
- Optical CATV networks
- CWDM and DWDM systems
- Passive optical networks
- Telecom, LANs

Features:

- Suitable to accommodate all types of adapters/connectors
- Modular patch panel design
- Color coded pigtails for ease of operation & maintenance
- Can be used for indoor & outdoor applications
- It is designed for splicing & patching options between pigtails & patch cords
- Options for single and multi-cable entries
- Easy installation and maintenance
- Optional rodent proof cover
- Option to terminate cable inside or outside the shelf

Rack Mount

The Rack Mount FMS is suitable for a 19"/21" standard rack. This type of FMS has easy access for splicing and maintenance due to sliding drawer design. Rack Mount FMS are available from 6 to 288 fiber configurations and can be customized as per customer requirements.



Specifications:

FMS Size	1U		2U		3U	4U	6U
Dimensions (H x W x D) mm	45 x 439 x 300		89 x 439 x 300		134 x 439 x 300	180 x 439 x 300	267 x 439 x 421
Adapter Type	Simplex	Duplex	Simplex	Duplex	Simplex	Simplex	Simplex
SC	6/12/24	48	48	96	48/72	96/144	288
LC	6/12/24	48	48	96	48/72	96/144	288
FC	6/12/24	48	48	96	48/72	96/144	288
Access	Front with pull-in drawer		Front with pull-in drawer		Front with pull-in drawer	Front with pull-in drawer	Front with pull-in drawer
Cable Dia. (mm)	8 to 18		8 to 18		8 to 18	8 to 20	8 to 20
Pigtail Length (mtr.)	1.5		1.5		1.5	1.5	1.5
Number of Splice Trays	Upto 2		Upto 6		6	8	12

Wall Mount

Wall Mount FMS Boxes provide mechanical & environmental protection for both fiber splicing and termination. It can either be used for only splicing of two fiber cables or splicing and patching with different types of connectors. This can be supplied with variants ranging from 6F to 144F splicing and termination.

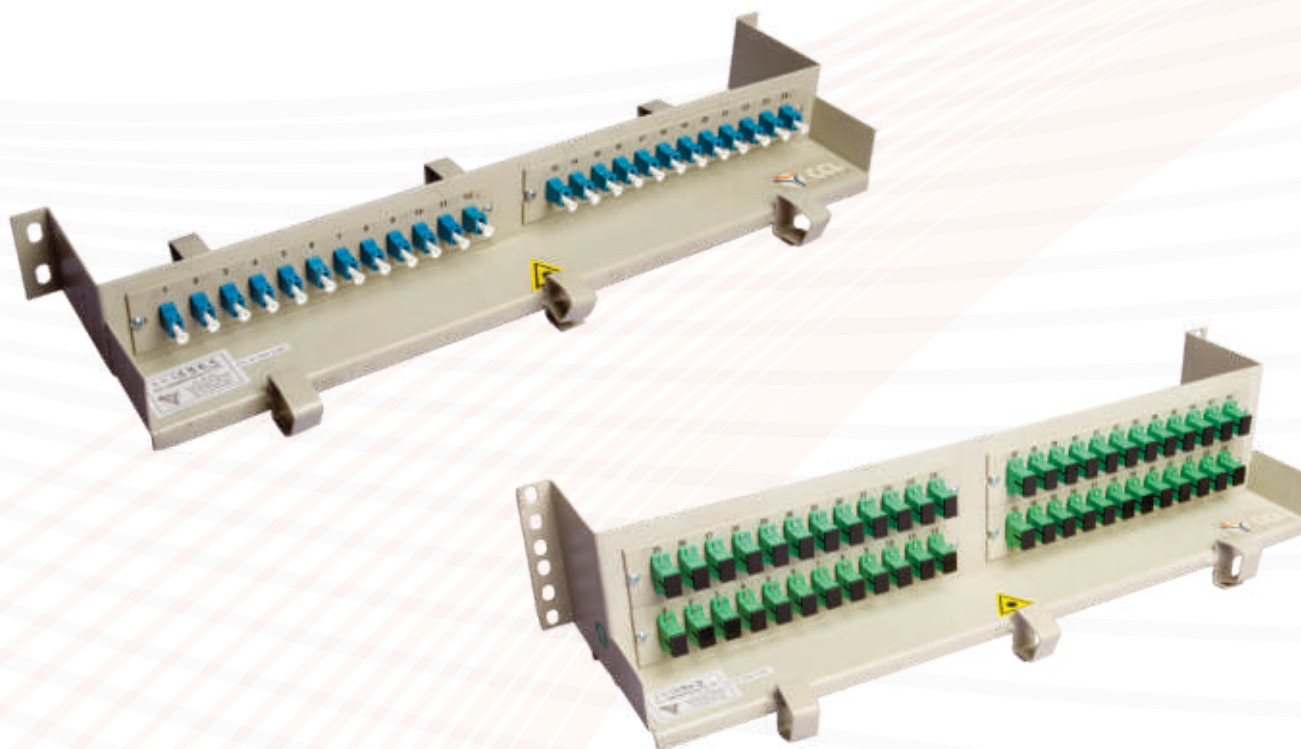


Specifications:

Parameters	Technical specifications		
Dimensions (H x W x D) mm	65 x 410 x 350	99 x 410 x 350	180 x 410 x 350
No. of adapters	6/12/24	48	96
No. of Splice trays	Max. 2	Max. 4	Max. 8
Maximum Splicing per tray	24	24	24

Optical Distribution Frames (ODF)

Optical Distribution Frames (ODF) are devices which act as a junction point to establish connection, interconnection & cross connection between communication equipment. This is normally designed for a rack mountable structure to suit 19"/21" for central office applications. The modular design of patch panels makes the system flexible to accommodate various types of adapters and the ODF can be customized as per customer requirements.



Specifications:

Dimension (H X W X D)	Maximum No. of Ports SM	Maximum No. of Ports DUP	Mounting
45 X 446 X 132mm	6F/12F/24F	48F	19" Rack mount
89 X 446 X 132 mm	48F	96F	19" Rack mount
133 X 446 X 132 mm	48F / 72F	-	19" Rack mount
178 X 446 X 132 mm	96F / 144F	-	19" Rack mount
267 X 446 X 132 mm	144F / 288F	-	19" Rack mount

Y-Cable Patch Panel

An external client protection module (that is, Y-Cable) is a passive device, and it can be inserted in 19-inch cabinet to implement 1+1 protection for 20 unidirectional client-side services (10 bi-directional client-side services). Based on the port types, Y-Cable modules are classified into single-mode and multi-mode modules.

Applications:

- FTTX (FTTP, FTTH, FTTN, FTTC)
- Passive Optical Networks (PON)
- Local Area Networks (LAN)
- CATV Systems
- Optical Signal Distribution
- Test Equipment

Features:

- Low Insertion loss
- Low PDL
- High Return Loss
- Uniform Power Splitting
- Compact Design
- Wide Operating Wavelength
- Wide Operating Temperature
- Excellent Environmental & Mechanical Stability
- Qualified Under Telcordia GR-1221 and GR-1209



Specifications:

Parameters	Unit	Technical Specification
Dimensions (H x W x D)	mm	45 X 439 X 222
Colour	-	Siemens Grey (7032)
Fiber Type	-	G657A1
Connector Type	-	LC-Duplex
Port Count	Nos	60
Split Ratio (COM-> P01/P02)	-	50:50
Operating Wavelength Range	nm	1260-1620
Operating Temperature	°C	-20 ~ +70
Storage Temperature	°C	-40 ~ +85
Insertion Loss (IL)	dB	≤ 0.4

IDF & MDF

An Intermediate Distribution Frame (IDF) & Main Distribution Frame (MDF) are a distribution frames in a central office or customer premises, which cross connects the user copper cable media to individual user line circuits and may serve as a distribution point for multipair copper cables from the Main Distribution Frame (MDF) or Intermediate Distribution Frame (IDF) to individual cables connected to equipment in areas remote from these frames. Often set up at a central office or on customer premises, an IDF/MDF helps cross-connect user copper cable media to individual user line circuits and serves as a distribution point for multipair cables from the Combined Distribution Frame (CDF) to separate cables connecting to devices located in different areas of the building.

Applications:

- Used for Telephone Exchange Central Office
- Customer-premises Equipment
- Wide Area Network
- Local Area Network

Features:

- The IDF/MDF are installed with back mounting frame and connection / disconnection modules
- Back mount frames are made of stainless steel
- Easy operation & maintenance
- Suitable for both indoor and outdoor applications

20 Pairs IDF Wall Mount Boxes

20 Pair Intermediate Distribution Frames (IDF) boxes are to be designed, fabricated & supplied made of CRCA sheet of 1mm thickness. These IDF boxes have the provision for mounting of connecting / disconnecting modules 10 pairs on SS back mounting frames fitted with panel lock & key on door & door sealing. The IDF boxes have the provision for cable tying and cable entries from the bottom. These boxes will be provided with wall mounting brackets and also earthing provision from outside.

Features:

- Capacity: 20 Pairs
- Supports Test and Connection Cords



Specifications:

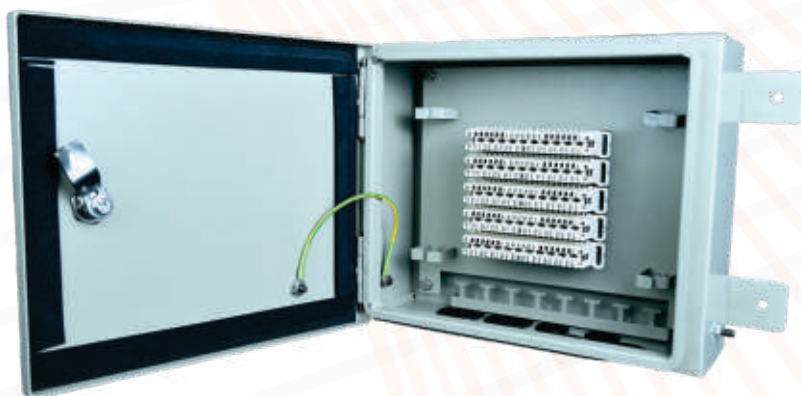
Sr. No.	Particulars	Specifications
1	Size (H x W x D)	175 x 225 x 95 (mm) +/- (5mm)
2	Thickness	1 mm, CRC +/- (0.2mm)
3	Colour	As per customer requirement
4	Back Mounting Frame	1 No. 20 Pair
5	Disconnection Module	2 Nos. of 10 Pair Each
6	Cable Entries	5 nos of 25 mm Dia Slot with Rubber Grommet
7	Accessories	Hinge, Lock, Key, Gasket & Fasteners
8	Sealing Protection	IP55

50 Pairs IDF Wall Mount Boxes

50 Pair Intermediate Distribution Frames (IDF) boxes are to be designed, fabricated & supplied made of CRCA sheet of thickness 1mm powder coated. These IDF boxes have the provision for mounting of connecting / disconnecting modules 10 pairs on SS back mounting frames fitted with panel lock & key on door & door sealing. These boxes will be supplied with 1 number of SS back mounting frame of 50 pairs and 5 nos of disconnection modules factory fitted. Disconnection module is of 10 pairs. The IDF boxes have the provision for cable tying and cable entries from the bottom. These boxes will be provided with wall mounting brackets and also earthing provision from outside.

Features:

- Capacity: 50 Pairs
- Supports following accessories
 - IPM protection wherever applicable
 - Labelling plates
 - Test and connection cords



Specifications:

Sr. No.	Particulars	Specifications
1	Size (H x W x D)	250 x 300 x 100 (mm) +/- (10mm)
2	Thickness	1 mm, CRC +/- (0.2mm)
3	Colour	As per customer requirement
4	Back Mounting Frame	1 No. 50 Pair
5	Disconnection Module	5 Nos. of 10 Pair Each
6	Cable Routing	Loops to be provided
7	Accessories	Hinge, Lock, Key & Gasket, Fasteners
8	Sealing Protection	IP55

250 Pair IDF & MDF Wall Mount Boxes

These Boxes have the provision for mounting of connecting / disconnecting modules 10 pairs on SS back mounting frames fitted with panel lock & key on door & door sealing. These boxes will be supplied with 3 nos of SS back mounting frames (2 nos of 100 pairs and 1 no of 50 pairs) and 25 nos of disconnection modules factory fitted. Disconnection module is of 10 pairs, each module will be provided with labelling plates wherever applicable. The capacity of these boxes is 250 pairs termination.

Features:

- Capacity: 250 Pairs
- Back mount frames of IDF/MDF are made of CRCA material
- Supports following accessories
 - IPM protection wherever applicable
 - Labelling plates
 - Test and connection cords



Specifications:

Sr. No.	Particulars	Specifications
1	Size (H x W x D)	570 x 500 x 280 (mm) +/- (20mm)
2	Thickness	1.2 mm, CRC +/- (0.2mm)
3	Colour	As per customer requirement
4	Back Mounting Frame	2 Nos. 100 Pair and 1 No. 50 Pair
5	Disconnection Module	25 Nos. of 10 Pair Each
6	IPM Module for Protection	As per customer requirement
7	Labeling Plates	2 Nos.
8	Cable Routing	Loops to be provided
9	Accessories	Hinge, Lock, Key & Gasket, Fasteners
10	Earthing Provision	Provided

1500 Pair IDF & MDF Wall Mount Boxes

These Boxes have the provision for mounting of connecting / disconnecting modules 10 pairs make SS back mounting frames fitted with panel lock & key on door & door sealing. These boxes will be fitted with 15 nos of SS back mounting frame and 150 nos of Connection disconnection modules factory fitted. Disconnection module is of 10 pairs, each module will be provided with labelling plates on top of each frame wherever applicable. The capacity of these boxes is 1500 pairs termination.

Features:

- Capacity: 250 Pairs termination
- Back mount frames of IDF/MDF are made of CRCA material
- MDF will be supplied with IPM wherever applicable



Specifications:

Sr. No.	Particulars	Specifications
1	Size (H x W x D)	1600 x 750 x 450 (mm) +/- (50mm)
2	Thickness	1.6 mm, CRC +/- (0.3mm)
3	Colour	As per customer requirement
4	Powder Coating	Structure Finish 80-100 micron
5	Back Mounting Frame	15 Nos of 100 Pair
6	Disconnection Module	150 Nos. of 10 Pair Each
7	IPM Module for Protection	As per customer requirement
8	Labelling Plates	Will be provided with each module
9	Cable Routing	Loops will be provided
10	Accessories	Hinge, Lock, Key & Gasket, Fasteners
11	Sealing Protection	IP55

Fiber Access Terminals

FAT - Fiber Access Terminal Boxes are devices used as a termination point for the feeder cables to connect drop cables in the FTTx network system. CCL Opto's FAT box is lightweight, compact and has controlled fiber routing guides. This box can accommodate splitters/cassettes & splices. FAT Boxes are suitable for both wall and pole mounting options.

Features:

- Plug in type splitter cassette (splitter quantities vary as per capacity)
- Integrated with splice cassette and cable management system
- Adapter Types: SC / PC & SC / APC
- IP 55 rated
- Compact and easy to install
- Suitable for both indoor and outdoor applications
- Wall and pole mounting options
- Accommodates different types of splitters
- Sizes available: 8, 16, 32, 64 Fibers

FAT 8F

Specifications:

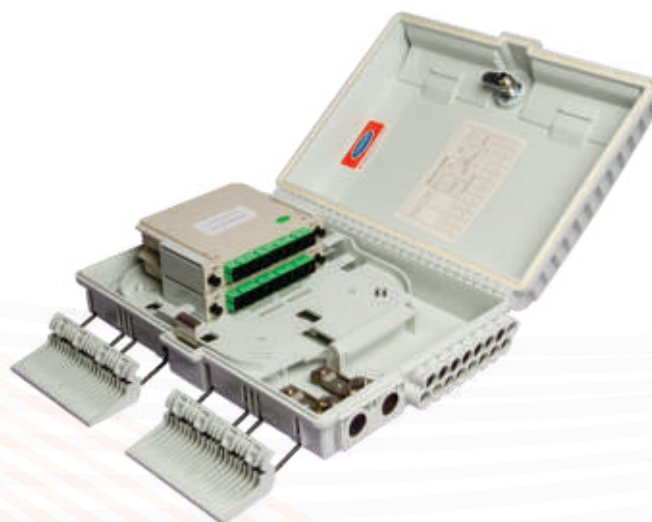
Parameters	Technical Specifications
Dimension (L x W x H)	252 X 200 X 75 mm
Material	PC + ABS
IP Rating	IP 55
Colour & Finishing	Grey
Adapter - Type Available	SC
Max. Capacity	Splice 8 fibers
PLC Splitter	1 Pcs of 1:8/2:8 Cassette splitter
Weight	0.96 Kgs



FAT 16F

Specifications:

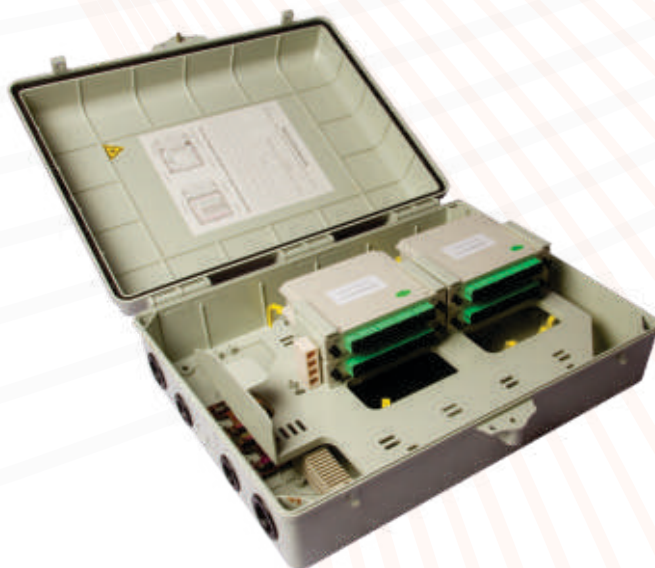
Parameters	Technical Specifications
Dimension (L x W x H)	300 X 230 X 85 mm
Material	PC + ABS
IP Rating	IP 55
Colour & Finishing	Grey
Adapter - Type Available	SC
Max. Capacity	Splice 16 fibers
PLC Splitter	1 Pcs of 1:16/2:16 Cassette splitter or 2 Pcs of 1:8/2:8 Cassette splitter
Weight	1.45 Kgs



FAT 32F

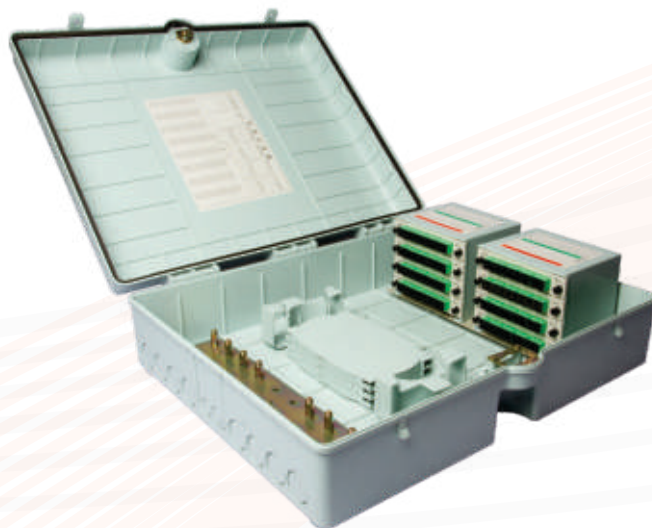
Specifications:

Parameters	Technical Specifications
Dimension (L x W x H)	380 X 320 X 110 mm
Material	PC + ABS
IP Rating	IP 55
Colour & Finishing	Grey
Adapter - Type Available	SC
Max. Capacity	Splice 32 fibers
PLC Splitter	4 Pcs of 1:8/2:8 Cassette splitter or 2 Pcs of 1:16/2:16 Cassette splitter
Weight	2.38 Kgs



Specifications:

Parameters	Technical Specifications
Dimension (L x W x H)	480 X 380 X 140 mm
Material	PC + ABS
IP Rating	IP 55
Colour & Finishing	Grey
Adapter - Type Available	SC
Max. Capacity	Splice 64 fibers
PLC Splitter	8 Pcs of 1:8/2:8 Cassette splitter or 4 Pcs of 1:16/2:16 Cassette splitter
Weight	3.55 Kgs



Optical Termination Boxes

Optical Termination Boxes (OTB) are a multi-purpose termination point in FTTx applications for end distribution network by providing connectivity (drop) to the subscriber. This box provides an option of splicing and patching in compact environments. It is compact in size with controlled fiber routing and drop cable management. This is suitable for wall-mount applications.

Features:

- Made of flame-retardant material, with age and ultraviolet resistant characteristics
- IP 55 rated
- Provision to store extra length of pigtails and drop cable
- Outlet of drop cable provided with special locking arrangement
- Area for splicing and termination are provided separately.

OTB 4F

Specifications:

Parameters	Technical Specifications
Dimensions (L x W x H)	180 x 110 x 40 mm
Material	PC + ABS
IP Rating	IP 55
Colour and Finishing	Light Grey
Adapter Types Available	SC
Maximum Splicing Capacity	4 Fibers
Weight	0.25 Kg
Mounting	Wall mount



OTB 6F

Specifications:

Parameters	Technical Specifications
Dimensions (L x W x H)	180 x 110 x 40 mm
Material	PC + ABS
IP Rating	IP 55
Colour and Finishing	Light Grey
Adapter Types Available	SC
Maximum Splicing Capacity	6 Fibers
Weight	0.25 Kg
Mounting	Wall mount



OTB 8F

Specifications:

Parameters	Technical Specifications
Dimensions (L x W x H)	200 x 170 x 50 mm
Material	PC + ABS
IP Rating	IP 55
Colour and Finishing	Light Grey
Adapter Types Available	SC
Maximum Splicing Capacity	8 Fibers
Weight	0.57 Kg
Mounting	Wall mount



OTB 12F

Specifications:

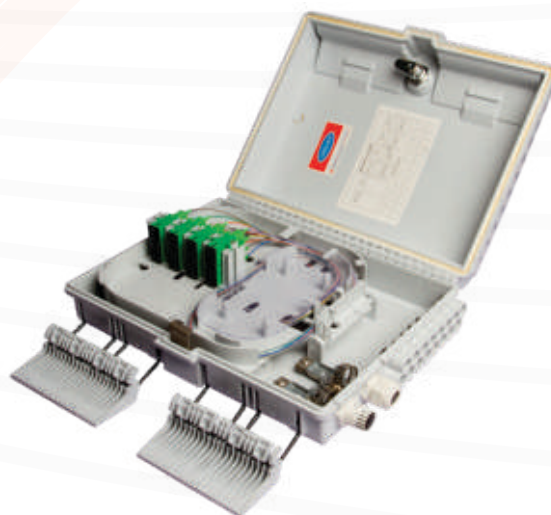
Parameters	Technical Specifications
Dimensions (L x W x H)	240 x 230 x 70 mm
Material	PC + ABS
IP Rating	IP 55
Colour and Finishing	Light Grey
Adapter Types Available	SC
Maximum Splicing Capacity	12 Fibers
Weight	0.745 Kg
Mounting	Wall mount



OTB 16F

Specifications:

Parameters	Technical Specifications
Dimensions (L x W x H)	280 x 175 x 75 mm
Material	PC + ABS
IP Rating	IP 55
Colour and Finishing	Light Grey
Adapter Types Available	SC
Maximum Splicing Capacity	16 Fibers
Weight	0.97 Kg
Mounting	Wall mount



OTB 24F

Specifications:

Parameters	Technical Specifications
Dimensions (L x W x H)	445 x 380 x 110 mm
Material	PC + ABS
IP Rating	IP 55
Colour and Finishing	Light Grey
Adapter Types Available	SC
Maximum Splicing Capacity	24 Fibers
Weight	1.31 Kg
Mounting	Wall mount



Specifications:

Parameters	Technical Specifications
Dimensions (L x W x H)	445 x 390 x 150 mm
Material	PC + ABS
IP Rating	IP 55
Colour and Finishing	Light Grey
Adapter Types Available	SC
Maximum Splicing Capacity	48 Fibers
Weight	3.12 Kg
Mounting	Wall mount



Home Termination Boxes (HTB)

Home Termination Boxes (HTB) are used in FTTx networks for the termination of optical fiber at the subscriber end. It has a provision to implement fiber splicing, protect fiber and provide optical ports for connecting to the subscriber.

Features:

- IP-45 rating
- Fiber bend radius of 30mm
- High quality industrial, anti-aging ABS plastic material.
- Wall mounting
- Suitable for FTTx indoor application
- Capacity of up to 2 port

Specifications:

Parameters	Details
Dimensions (mm)	85*85 (±1)
Adapter Type Available	SC/PC & SC/APC
Material	ABS
Nos. Adapter SC/PC Or SC/APC	2
Mounting	Wall mount



Fiber Distribution Cabinets (FDC)

Fiber Distribution Cabinets (FDC) are used in optical fiber network to interconnect, cross connect and distribution of optical fiber links and power. It is suitable for both splicing and termination of optical fiber cables and splitting of fiber with the help of splitters. It is very easy to install, operate and maintain.

FDC is made of glass fiber reinforced unsaturated polyester plastic for outdoor application with protection rating up to IP55.

Environmental Specification	
External temperature of cabinet	~ -40°C +60°C
Environmental temperature	~ -30°C +55°C
Working environment temperature of equipment	~ -18°C +35°C
Environment humidity	≤95%
Atmospheric pressure in +30°C	70~106kpa



Specifications:

Parameters	96F	144F
Dimension (L x W x H)	445 x 270 x 830 mm	548 x 305 x 1257 mm
No. of Splice Tray	8	12
Weight KG	20Kg Unloaded, 22 ~ 24 Kg (Loaded)	28.40Kg Unloaded, 32 ~ 33 Kg (Loaded)
Material	SMC	SMC
Installation	Floor Mounts Applications	Floor Mounts Applications

Riser Cables

Riser Cables are very compact and flexible cable construction for fast installation in existing or new building conduits for FTTx application. It consists of tight buffer optical cable ranges from 2 to 24 core 900µm tight buffer fiber as an optical communication medium.

The tight buffer fiber is wrapped with a layer of aramid yarn which act as strength member and outer jacket is made of LSZH/OFNR material.



Specifications:

Cable Count	Outside diameter	Tight buffer diameter	Weight	Minimum allowable tensile strength (N)		Minimum allowable crush load (N/100mm)		Minimum bending radius (MM)		Storage temperature
	(MM)	(MM)	(KG)	Short term	Long term	Short term	Long term	Short term	Long term	(°C)
12	6	0.9	35	1000	500	1000	200	20D	10D	30
24	8.3	0.9	55	1000	500	1000	200	20D	10D	30

Drop Cables

The flat type Drop Cable is normally used for FTTx end connectivity from S2 splitting stage to subscriber end. This is designed for aerial, duct or conduit application. The dielectric strength member (FRP) running parallel to optical fiber provides longitudinal strength.

Specifications:

Parameter	Description	Construction
Strength member	Nos.	0.5mm steel wire
Fiber type	TELCORDIA	SM G657A1
Fiber Count	Nos.	1 or 2
Fiber Color	Color	Blue, Orange
Cable Diameter	H x B	2.0 x 3.0 MM
Cable color	color	White or Black



Fast Assembly Connectors

Field Assembly Connectors are also known as Fast Connectors are widely used in FTTx applications. These connectors are normally used to connect 1F or 2F drop cable from S2 splitter port to ONT at subscriber end. These connectors are reliable and fast to install with low losses.

Specifications:

Parameter	Technical Specifications
Application scope	3.1*2.0mm butterfly optical cable
Application model	Single model
Insertion loss	≤ 0.3 dB 1310nm & 1550nm
Return loss	≥ 50 dB
Success rate	> 98%
Assembly times	> 10 times
Tensile strength	> 50N
Online stretching resistance test (30N)	Δ IL ≤ 0.3 dB
Mechanical permanence (500 times)	Δ IL ≤ 0.3 dB
Drop test(4m high concrete floor/each direction one time/total three times	Δ IL ≤ 0.3 dB



OFC ACCESSORIES

Fiber Optic Patch Cords

CCL Opto's Patch Cords are passive telecommunication components used for interconnection of optical equipment to optical fiber cables. They are single fiber cables pre-assembled with a connector at both ends. This is also known as a jumper cable and has connectors at both ends (either same or different types). The optical fiber patch cords are designed for use in telecommunications, CATV, data communications and local area network used for both indoor and outdoor applications.

Applications:

- Patch Cords are designed to provide interconnection solutions for Telecommunication Network
- CATV System
- Local Area Network
- Active / Passive devices
- FTTx Applications

Versions Available:

- Single mode or Multimode
- Simplex or Duplex
- 900um, 2mm & 3mm
- Armoured or Unarmoured
- Connectors - SC, LC, FC, ST, MU, MTRJ, E2000 etc.
- Available in all type of Connector - PC, APC, UPC
- Cable options - PVC, LSZH



Specifications:

Parameters	Unit	Description					
		SC, LC, FC, ST, MU				E2000(HS)	
		PC		APC		PC	APC
		SM	MM	SM	MM	SM	SMC
Insertion loss(IL)	dB	≤ 0.2	≤ 0.2	≤ 0.2	≤ 0.2	≤ 0.15	≤ 0.15
Return loss (RL)	dB	≥ 50	≥ 25	≥ 65	≥ 40	≥ 50	≥ 85
Repeatability	dB	≤ 0.2					
Changeability	dB	≤ 0.2					
Operating temperature	$^{\circ}\text{C}$	-35 to $+70$					
Storage temperature	$^{\circ}\text{C}$	-35 to $+70$					
Fiber type	-	G657(A1/A2) / G652D					
Fiber diameter	um	SM-9 ,OR MM-50/125(OM2,OM3,OM4) and 62.5/125(OM1)					

MPO Patch Cords

MPO (Multi-fiber Push On) are connectors for ribbon cables with four to twenty-four fibers connectors for single mode fiber and have angled ends to minimize back-reflection, while multimode fiber versions typically have flat ends. MPO Cassette makes it easy to switch the connection between a multi-fiber MPO connector and single-fiber connector. This system is designed for a high-density termination environment in telecom central offices, data centers etc.



Specifications:

Item		Unit	Parameter			
MPO/MTP Connector			Standard SM MT Ferrule	Low Loss (Elite) SM MT Ferrule	Standard MM MT Ferrule	Low Loss MM MT Ferrule
Insertion loss(IL)	Typical Maximum	dB	0.35	0.20	0.35	0.20
		dB	0.75	0.35	0.60	0.35
Return Loss (RL)		dB	≥ 60 for APC (8 Angle Polish)		≥ 20 for PC	
		dB	≥ 50 for PC		-	
Male/Female		dB	Male: With Pins, Female: Without Pins			
Tensile Strength		dB	≤ 0.3 (Max 66 N)			
Vibration		dB	≤ 0.3 (10 ~ 55 Hz)			
Thermal Cycle		dB	≤ 0.3 (-40 ~ 75 °C, 21 Cycles)			

Fiber Optic Splitters

We offer several types of high-quality splitters. These products are designed for long-term reliability, superior performance and a wide range of use in fiber optics including optical transmission, long haul, subscriber loop and fiber to the home, telecommunications, CATV, test equipment, optical fiber sensors, data communications, and wide-area network. Our optical splitters are designed for consistent performance, low optical loss, low polarization sensitivity, superior environmental and mechanical characteristics, and fast installation. They also enable more effective monitoring and management of fiber optical networks.

Types of Splitters:



- Tube type



- ABS box



- LGX/ Cassette type

Specifications:

1×N (N≥2) PLC Splitter							
Operating Wavelength (nm)	1260 - 1650						
Port configuration	1 x 2	1 x 4	1 x 8	1 x 16	1 x 32	1 x 64	1 x 128
Insertion loss (dB)	3.8/4.0	7.1/7.3	10.2/10.5	13.5/13.7	16.5/16.9	20.5/21.0	23.8/24.2
Loss uniformity (dB)	0.4	0.6	0.8	1.2	1.5	2.0	2.5
Polarization dependent loss (dB)	0.2	0.2	0.2	0.25	0.3	0.35	0.4
Wavelength dependent loss (dB)	0.3	0.3	0.3	0.5	0.5	0.5	0.5
Temperature dependent loss (-40 to 85) (dB)	0.4	0.4	0.4	0.5	0.5	0.5	0.5
Return loss(dB)	≥45 (PC); ≥55 (APC)						
Operating temperature	-40~+85						
Directivity (dB)	≥ 55						
Storage temperature	-40~+85						

Specifications:

2×N (N≥2) PLC Splitter						
Operating Wavelength (nm)	1260 - 1650					
Port configuration	2 x 2	2 x 4	2 x 8	2 x 16	2 x 32	2 x 64
Insertion loss (dB)	≤ 4.4	≤ 7.7	≤ 10.8	≤ 14.1	≤ 17.4	≤ 20.7
Loss uniformity (dB)	≤ 0.8	≤ 0.8	≤ 1.0	≤ 1.5	≤ 1.8	≤ 2.0
RL (dB)	≥45 (PC); ≥55 (APC)					
Polarization dependent loss (dB)	≤ 0.2	≤ 0.2	≤ 0.3	≤ 0.3	≤ 0.3	≤ 0.4
Humidity	≤ 95% (+40℃)					
Operating temperature	-40~+85℃					
Directivity (dB)	≥ 55					
Storage temperature	-40~+85℃					
Atmospheric pressure	62~106kPa					

Fiber Optic Attenuators

Fiber Optic Attenuators are used in fiber optic network communication systems to reduce optical power in fiber links. In general, attenuators are used in single-mode long-haul applications to prevent optical overload at the receiver end. These high-performance devices are designed to give accurate attenuation over a wide range of wavelengths.



Features:

- Low insertion loss and back reflection
- Definable attenuation from 1dB to 30dB
- Types - PC,UPC, and APC polish
- Available with FC, SC, ST, LC etc. type connectors

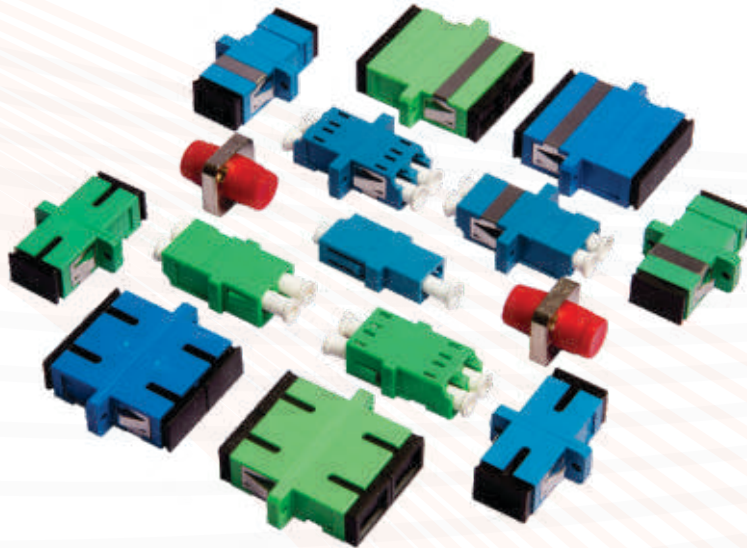
Specifications:

Adapter type Fixed Attenuator	
Attenuator Range	1-30 dB
Available Wavelengths	1310 nm or 1550 nm
Fixed Attenuation Value	1,2,3,5,10,15,20 dB or Optional
Return Loss	≥ 50 dB (SPC)
	≥ 60 dB (APC)
Attenuation Accuracy	+/-0.5 (1-5) dB
	+/- 10% (6-30) dB
Polarization Dependent Loss	≤ 0.2 dB
Humidity Range	0.2 dB change in 10% to 90% relative humidity range
Vibration	≤ 0.1 dB change between 10 Hz to 55 Hz
Drop	0.2 dB after 8 drops (3 axes) from 1.8 meters onto a hard surface.

Plug-in Fixed Attenuator	
Operating Wavelength	SM : 1200-1600 nm or 1310 nm, 1550 nm
	MM : 850 nm, 1300 nm
Return Loss	≥ 50 dB (UPC)
	≥ 60 dB (APC)
Attenuation Accuracy	+/-0.5 (1-5) dB
	+/- 10% (6-30) dB
Polarization Dependent Loss	≤ 0.2 dB
Maximum Optical Input Power	200 mW
Operating Temp. Range	-40° C ~ 80° C

Fiber Optic Adapters

Fiber Optic Adapters or Couplers are designed and manufactured to control refractive index of two connection points of optical fiber cables. This connection is mechanical in nature and highly durable, performance oriented, reliable and has the low insertion loss in telecom, metro and access applications.



Features:

- Low Insertion loss
- Highly precision alignment
- Compact design
- With or without flange

Types Available:

- SC / PC, SC / APC, LC / PC, LC / APC
- FC / PC, FC / APC
- MU / PC, MU / APC
- E2000 / PC, E2000 / APC

Specifications:

Parameters	Technical Specifications
Insertion Loss	< 0.1 dB $\pm$ 0.05dB
Durability	0.05 dB typical change, 500 mating

PRODUCTION FACILITIES



Patch Cords Production



Splice Joint Closures Production



Patch Cords Crimping



Fiber Management Systems Production

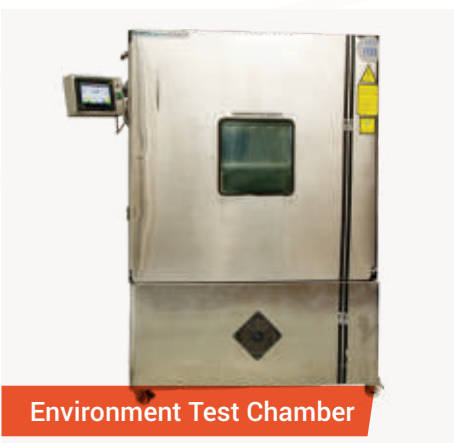
TESTING FACILITIES



Patch Cord Testing



UV Weather Testing Chamber



Environment Test Chamber



Splice Joint Closure Testing

ABBREVIATIONS

Sr. No	Short Form	Long Form
1	SJC	Splice Joint Closure
2	HS	Heat Shrink
3	IP	Ingress Protection
4	UV	Ultra-Violet
5	PPCP	Poly Propylene Co-Polymer
6	ABS	Acrylonitrile-Butadiene-Styrene
7	FMS	Fiber Management System
8	FTTH	Fiber to the Home
9	CWDM	Coarse Wavelength Division Multiplexing
10	DWDM	Dense Wavelength Division Multiplexing
11	CATV	Community Access Television
12	LAN	Local Area Network
13	RM	Rack Mount
14	WM	Wall Mount
15	SC	Square Connector
16	LC	Lucent Connector
17	FC	Ferrule Connector
18	ODF	Optical Distribution Frame
19	FAT	Fiber Access Terminals
20	PC	Physical Contact
21	APC	Angled Physical Contact
22	PC + ABS	Poly-Carbonate + Acrylonitrile Butadiene Styrene
23	OTB	Optical Termination Box
24	FDC	Fiber Distribution Cabinet
25	LSZH	Low Smoke Zero Halogen
26	OFNR	Optical Fiber Nonconductive Riser
27	FRP	Fibre-reinforced Plastic
28	SM	Single Mode
29	MM	Multi-Mode
30	MPO	Multi-fiber Push On
31	PLC Splitter	Planar Lightwave Circuit Splitter
32	IL	Insertion Loss
33	RL	Return Loss



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